Healthcare Solutions on Snowflake

Cortex Code Skills & Profile Architecture Guide

A modular, skill-based approach to building healthcare data solutions using Snowflake platform capabilities

Built with Cortex Code | March 2026

Table of Contents

1. Executive Summary

This document describes a modular, skill-based architecture for building healthcare data solutions on Snowflake using Cortex Code. The system consists of:

- 15 healthcare domain skills covering 7 business functions
- 1 orchestrator agent profile (healthcare-solutions) that routes requests to the right skill(s)
- Integration with Snowflake platform skills (Dynamic Tables, Cortex AI, Streamlit, dbt, governance)
- 2 Cortex Knowledge Extensions (CKEs) for PubMed biomedical literature and ClinicalTrials.gov research
- 1 Data Model Knowledge Repository (Cortex Search over DICOM 18-table reference model)
- 9 cross-domain composition patterns for complex multi-skill solutions
- Cross-cutting prerequisite pattern: auto-fires data model queries before schema-dependent tasks

The architecture enables a healthcare solutions architect to quickly build end-to-end solutions by composing domain-specific skills with Snowflake platform capabilities, while enforcing HIPAA governance guardrails across all workflows.

Key Benefits

- **Modularity:** Reusable skills that encode domain expertise and best practices
- **Intelligent Routing:** Single orchestrator routes to the right skill based on natural language intent
- **Platform Integration:** Healthcare skills invoke Snowflake platform skills for infrastructure
- **Composability:** Cross-domain patterns compose multiple skills for complex use cases
- **Governance by Default:** HIPAA guardrails enforced as cross-cutting concerns
- **Knowledge-Grounded:** CKEs provide RAG-based biomedical evidence; Cortex Search grounds schemas in live data model definitions

2. Architecture Overview

The architecture follows a four-layer model: external knowledge tools (CKEs) at the top, an orchestrator agent profile, healthcare domain skill collections, and Snowflake platform skills at the foundation.

Layered Architecture

```
+-----------------------------------------------------------------+
|              CORTEX KNOWLEDGE EXTENSIONS (CKEs)                 |
|  PubMed Biomedical Corpus  |  Clinical Trials Research Database  |
+-----------------------------------------------------------------+
                              |
+-----------------------------------------------------------------+
|              ORCHESTRATOR AGENT PROFILE                          |
|              healthcare-solutions.md                             |
|  Intent Detection -> Domain Routing -> Skill Composition        |
+-----------------------------------------------------------------+
       |              |              |              |
       v              v              v              v
+-------------+ +------------+ +------------+ +------------+
| Medical     | | Clinical   | | Drug       | | Genomics & |
| Imaging     | | Data / EHR | | Safety     | | Bioinform. |
| (7 sub-     | | (3 skills) | | (2 skills) | | (5 skills) |
| skills)     | |            | |            | |            |
+-------------+ +------------+ +------------+ +------------+
       |     +----------+ +----------+ +----------+      |
       |     | Claims & | | Lab Data | | Research |      |
       |     | RWE (1)  | | (1)      | | (1)      |      |
       |     +----------+ +----------+ +----------+      |
       v              v              v              v
+-----------------------------------------------------------------+
|              SNOWFLAKE PLATFORM SKILLS                           |
| Dynamic Tables | Cortex AI | Streamlit | SPCS | dbt | ML        |
| React App | Governance | Cortex Search | Cortex Agent | Analyst  |
+-----------------------------------------------------------------+
       ^
       |
+-----------------------------------------------------------------+
|         DATA MODEL KNOWLEDGE REPOSITORY                         |
| Excel -> CSV -> Table -> Cortex Search Service                  |
| DICOM_MODEL_SEARCH_SVC (auto pre-step for schema tasks)         |
+-----------------------------------------------------------------+
```

How It Works

Step	Action	Component
1	User sends a natural language request	Cortex Code CLI
2	Orchestrator detects healthcare business domain from trigger keywords	healthcare-solutions.md

Step	Action	Component
3	Matching domain skill is invoked (e.g., $healthcare-imaging for DICOM)	Domain Skill Router
4	For schema-dependent intents (PARSE, INGEST, ANALYTICS, GOVERNANCE), auto-query Data Model Knowledge Repository via Cortex Search	DICOM_MODEL_SEARCH_SVC
5	Domain skills may invoke CKEs (PubMed, Clinical Trials) for evidence grounding	CKE Cortex Search Services
6	Domain skills invoke Snowflake platform skills for infrastructure	Dynamic Tables, Streamlit, etc.
7	HIPAA governance guardrails applied as cross-cutting concerns	Masking, RLS, Audit

3. Healthcare Skills Inventory

The collection contains 15 standalone skills organized into 7 healthcare business functional area categories in the repository, plus 1 skill collection (healthcare-imaging) with 7 sub-skills.

Repository Directory Structure

```
skills/
+-- medical-imaging/          healthcare-imaging (router + 7 sub-skills), dicom-parser
+-- clinical-data-ehr/        fhir-data-transformation, clinical-nlp, omop-cdm-modeling
+-- drug-safety/              pharmacovigilance, clinical-trial-protocol-skill
+-- claims-rwe/               claims-data-analysis
+-- genomics-bioinformatics/  nextflow-development, variant-annotation, single-cell-rna-qc,
|                             scvi-tools, survival-analysis
+-- lab-instrument-data/      instrument-data-to-allotrope
+-- research-strategy/        scientific-problem-selection
```

3.1 Medical Imaging & Radiology

Repo path: skills/medical-imaging/

The healthcare-imaging skill collection is a router that detects intent and routes to 7 sub-skills covering the full imaging lifecycle from parsing to AI, plus a data model knowledge service.

Sub-Skill	Triggers	Description
dicom-parser	DICOM parse, extract tags, pydicom, DICOM data model	18-table DICOM data model + pydicom parser script
dicom-ingestion	Ingest DICOM, imaging pipeline, load images, stage	Stages, COPY, Dynamic Tables, Streams/Tasks
dicom-analytics	Imaging analytics, radiology NLP, report extraction	Cortex AI NLP on reports, Cortex Search, study metrics
imaging-viewer	Imaging viewer, Streamlit imaging, DICOM viewer	Streamlit dashboard + SPCS pixel viewer
imaging-governance	HIPAA imaging, PHI masking, imaging audit	Masking policies, classification, row-access, audit
imaging-ml	Imaging model, pathology model, radiology AI	ML training, Model Registry, SQL inference
data-model-knowledge	Data model reference, DICOM schema lookup, PHI columns	Cortex Search over 18-table DICOM model (auto pre-step)

3.2 Clinical Data & EHR

Repo path: skills/clinical-data-ehr/

Skill	Triggers	Description
fhir-data-transformation	FHIR, HL7, Patient, Observation, Bundle, ndjson	Transforms FHIR R4 resources into analytics-ready relational tables
clinical-nlp	Clinical NLP, NER, clinical notes, ICD coding, med extraction	Extracts structured data from clinical text via Cortex AI / spaCy
omop-cdm-modeling	OMOP, CDM, OHDSI, vocabulary mapping, SNOMED, LOINC	Transforms EHR/claims to OMOP CDM v5.4 with vocab mapping

3.3 Drug Safety & Pharmacovigilance

Repo path: skills/drug-safety/

Skill	Triggers	Description
pharmacovigilance	FAERS, adverse events, drug safety, MedDRA, PRR, ROR	FDA FAERS analysis for drug safety signal detection
clinical-trial-protocol-skill	Clinical trial protocol, design study, FDA submission	Generates clinical trial protocols via waypoint architecture

3.4 Claims & Real-World Evidence

Repo path: skills/claims-rwe/

Skill	Triggers	Description
claims-data-analysis	Claims, RWE, 837/835, utilization, HEDIS, PDC	Cohort building, utilization metrics, treatment patterns, adherence

3.5 Genomics & Bioinformatics

Repo path: skills/genomics-bioinformatics/

Skill	Triggers	Description
nextflow-development	nf-core, Nextflow, FASTQ, variant calling, GEO, SRA	Runs nf-core pipelines (rnaseq, sarek, atacseq) on sequencing data
variant-annotation	VCF, ClinVar, gnomAD, pathogenic variants, ACMG	Annotates genomic variants with pathogenicity and frequency data
single-cell-rna-qc	scRNA-seq, QC, scanpy, MAD-based filtering	Automated QC for single-cell RNA-seq using scverse best practices
scvi-tools	scVI, scANVI, totalVI, batch correction, integration	Deep learning single-cell analysis using scvi-tools VAE models
survival-analysis	Kaplan-Meier, Cox regression, hazard ratio, PFS, OS	Time-to-event analysis with publication-ready plots

3.6 Lab & Instrument Data

Repo path: skills/lab-instrument-data/

Skill	Triggers	Description
instrument-data-to-allotrope	Instrument files, Allotrope, ASM, LIMS, ELN	Converts lab instrument outputs to Allotrope Simple Model JSON/CSV

3.7 Research Strategy

Repo path: skills/research-strategy/

Skill	Triggers	Description

Skill	Triggers	Description
scientific-problem-selection	Research problem, project ideation, evaluate project	Systematic problem selection via Fischbach & Walsh decision trees

4. Orchestrator Agent Profile

The healthcare-solutions agent profile is a custom Cortex Code agent defined as a markdown file at ~/.snowflake/cortex/agents/healthcare-solutions.md. It serves as the single entry point for all healthcare tasks, detecting intent and routing to the appropriate skill(s).

Profile Structure

```
---
name: healthcare-solutions
description: "Healthcare industry solutions architect for Snowflake.
  Orchestrates skills across medical imaging, clinical data, drug safety,
  claims/RWE, genomics, and lab data. Integrates CKEs for PubMed and
  ClinicalTrials.gov research."
tools: ["*"]
---
# Healthcare Solutions Profile
You are a Healthcare Solutions Architect specializing in building
end-to-end healthcare data solutions on Snowflake.
## Skill Routing         [Intent detection tables for 7 business domains]
## CKE Tools             [PubMed + Clinical Trials search integration]
## Cross-Domain Patterns [9 composition patterns for multi-skill solutions]
## Guardrails            [HIPAA governance rules across all workflows]
```

Routing Mechanism

The profile uses trigger keywords in the user's request to determine the healthcare business domain. Each domain maps to one or more skills via $skill-name invocation syntax. CKEs are automatically suggested when evidence grounding is beneficial.

Domain	Example Request	Skill Invoked	Platform / CKE
Medical Imaging	"Parse these DICOM files"	$healthcare-imaging	Dynamic Tables, SPCS
Clinical / EHR	"Transform FHIR bundles"	$fhir-data-transformation	Dynamic Tables, dbt
Drug Safety	"Analyze FAERS for drug X"	$pharmacovigilance	Cortex AI, PubMed CKE
Claims / RWE	"Build RWE cohort"	$claims-data-analysis	Cortex Analyst, Trials CKE
Genomics	"Annotate VCF variants"	$variant-annotation	ML Registry
Lab Data	"Convert to Allotrope"	$instrument-data-to-allotrope	Dynamic Tables
Research	"Evaluate this research idea"	$scientific-problem-selection	PubMed CKE

5. Cortex Knowledge Extensions (CKEs)

Two Cortex Knowledge Extensions from the Snowflake Marketplace are available as RAG-based knowledge tools. These are shared Cortex Search Services that provide domain-specific literature search without copying data into your account.

Available CKEs

CKE	Marketplace Listing	Service Name	Use Cases
PubMed Biomedical Research Corpus	GZSTZ67BY9OQW	<CKE_DB>.SHARED.CKE_PUBMED_SERVICE	Literature review, drug mechanisms, radiology research, clinical NLP context
Clinical Trials Research Database	GZSTZ67BY9ORD	<CKE_DB>.SHARED.CKE_CLINICAL_TRIALS_SERVICE	Trial design, protocol comparison, feasibility analysis, eligibility criteria

CKE Setup (One-Time)

1. Navigate to Snowflake Marketplace and search for the CKE listing.
2. Click Get to install - no data is copied; a shared Cortex Search Service appears in your account.
3. Note the database name assigned (e.g., PUBMED_BIOMEDICAL_RESEARCH_CORPUS).

CKE Query Pattern (SQL)

```
SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
  '<cke_database>.SHARED.<service_name>',
  '{"query": "<natural language question>",
    "columns": ["chunk", "document_title", "source_url"]}'
);
```

CKE with Cortex Agent API

CKEs can be specified as cortex_search tools in the Cortex Agent API:

```
{
  "tools": [{
    "tool_spec": {
      "type": "cortex_search",
      "name": "pubmed_search",
      "spec": {
        "service_name": "<cke_database>.SHARED.CKE_PUBMED_SERVICE",
        "max_results": 5,
        "title_column": "document_title",
        "id_column": "source_url"
      }
    }
  }]
}
```

CKE Routing by Skill

CKE	Trigger Keywords	Skills

CKE	Trigger Keywords	Skills
PubMed CKE	PubMed, biomedical literature, drug mechanism, clinical evidence	$pharmacovigilance, $clinical-nlp, $scientific-problem-selection, dicom-analytics
Clinical Trials CKE	ClinicalTrials.gov, trial search, trial design, feasibility	$clinical-trial-protocol-skill, $claims-data-analysis, $survival-analysis

Integration Example: Drug Safety + PubMed CKE

```
WITH faers_signals AS (
  SELECT drug_name, reaction_pt, prr, ror
  FROM drug_safety_signals WHERE prr > 2 AND ror > 2
),
literature_evidence AS (
  SELECT s.drug_name, s.reaction_pt, s.prr,
    SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
      '<CKE_DB>.SHARED.CKE_PUBMED_SERVICE',
      '{"query": "' || s.drug_name || ' ' || s.reaction_pt
        || ' adverse event mechanism",
        "columns": ["chunk", "document_title", "source_url"]}'
    ) AS pubmed_evidence
  FROM faers_signals s
)
SELECT * FROM literature_evidence;
```

6. Data Model Knowledge Repository

Reference data models are stored as searchable knowledge repositories on Snowflake using Cortex Search. Skills query the repository at runtime to get the latest table/column definitions instead of relying on hardcoded schemas.

Architecture

```
Excel Spreadsheet (dicom_data_model_reference.xlsx)
    |
    v  export_search_corpus_csv.py
CSV (dicom_model_search_corpus.csv)      222 rows, 18 tables
    |
    v  COPY INTO
Snowflake Table (DICOM_MODEL_REFERENCE)
    |
    v  Cortex Search Service
DICOM_MODEL_SEARCH_SVC  <-- Skills query this at runtime
```

DICOM Model: 18 Tables, 222 Columns

The DICOM data model covers the full imaging hierarchy plus supporting entities:

Category	Tables	Purpose
Core Hierarchy	dicom_patient, dicom_study, dicom_series, dicom_instance, dicom_frame	Patient -> Study -> Series -> Instance -> Frame
Technical Context	dicom_equipment, dicom_image_pixel, dicom_image_plane	Scanner/device details, pixel encoding, spatial positioning
Workflow	dicom_procedure_step	Requested/performed procedure mapping
Dose	dicom_dose_summary	CT/radiography exposure parameters
Derived Objects	dicom_segmentation_metadata, dicom_structured_report_header	SEG and SR SOP metadata
Storage	dicom_file_location	Physical file location and access URIs
Generic Elements	dicom_element, dicom_sequence_item	Long-tail tags and sequence items
ML / Embeddings	image_embedding, embedding_model, embedding_evaluation	Vector representations and model catalog

Cross-Cutting Prerequisite (Auto Pre-Step)

For DICOM-related intents (PARSE, INGEST, ANALYTICS, GOVERNANCE), the healthcare-imaging router automatically executes Step 0 before dispatching to the sub-skill. This grounds all schema work in the latest data model.

```
SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
    'UNSTRUCTURED_HEALTHDATA.DATA_MODEL_KNOWLEDGE.DICOM_MODEL_SEARCH_SVC',
    '{"query": "<context from user request>",
      "columns": ["table_name", "column_name", "data_type",
                  "constraints", "description", "dicom_tag",
                  "contains_phi", "relationships"]}'
);
```

Intent	Step 0 Query Focus	What Gets Grounded
PARSE	All tables + columns for requested scope	CREATE TABLE DDL statements
INGEST	Target table columns + data types + relationships	COPY INTO mappings, Dynamic Table SELECT lists
ANALYTICS	Source table columns + descriptions	Analytical views, Cortex AI extraction prompts
GOVERNANCE	PHI-flagged columns across all tables	Masking policy targets, de-identification scope

DDL Generation from Search Results

```
WITH model_knowledge AS (
    SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
        'UNSTRUCTURED_HEALTHDATA.DATA_MODEL_KNOWLEDGE.DICOM_MODEL_SEARCH_SVC',
        '{"query": "dose summary radiation CT exposure",
          "columns": ["table_name","column_name","data_type","constraints"]}'
    ) AS context
)
SELECT SNOWFLAKE.CORTEX.COMPLETE(
    'llama3.1-70b',
    'Generate Snowflake CREATE TABLE DDL from this data model reference. '
    || 'Use exact column names, data types, and constraints. Reference: '
    || context::STRING
) AS generated_ddl
FROM model_knowledge;
```

Extending to Other Data Models

Domain	Spreadsheet	Search Service
DICOM Imaging	dicom_data_model_reference.xlsx	DICOM_MODEL_SEARCH_SVC (live)
FHIR R4	fhir_r4_resource_model.xlsx	FHIR_MODEL_SEARCH_SVC (planned)
OMOP CDM v5.4	omop_cdm_v54_model.xlsx	OMOP_MODEL_SEARCH_SVC (planned)
FAERS	faers_data_model.xlsx	FAERS_MODEL_SEARCH_SVC (planned)
Claims (837/835)	claims_data_model.xlsx	CLAIMS_MODEL_SEARCH_SVC (planned)

7. Snowflake Platform Skills Integration

Healthcare domain skills leverage Snowflake's bundled platform skills for infrastructure, compute, AI, governance, and visualization.

7.1 Data Engineering

Dynamic Tables

Used for incremental data pipelines with automatic refresh:

- **DICOM metadata ingestion:** Dynamic Tables with 10-minute TARGET_LAG for continuous imaging data refresh
- **FHIR resource flattening:** Incremental FHIR JSON bundles into relational tables
- **Imaging analytics:** Study volume metrics refreshed every 30 minutes
- **Claims data:** Incremental claims aggregation

Streams & Tasks

- **DICOM ingestion:** Stream on raw landing table triggers processing of new imaging metadata
- **HL7v2 messages:** Stream on raw HL7 triggers ADT event parsing

dbt Projects

- **OMOP CDM:** dbt for vocabulary mapping and CDM table materialization
- **Claims analytics:** dbt for cohort tables, utilization summaries, HEDIS measures

7.2 AI & Machine Learning

Cortex AI Functions

- **EXTRACT_ANSWER:** dicom-analytics extracts findings/impressions from radiology reports
- **SUMMARIZE:** Condenses lengthy radiology reports for dashboards
- **COMPLETE:** clinical-nlp uses LLM for complex entity extraction; DDL generation from model search
- **SENTIMENT:** Flags potentially critical radiology findings

Cortex Search

- **Imaging metadata search:** Semantic search over studies/reports (e.g., 'chest CT with pulmonary nodule')
- **Data model knowledge:** DICOM_MODEL_SEARCH_SVC provides live schema definitions to skills at runtime
- **CKE integration:** PubMed and Clinical Trials shared search services for evidence grounding

ML Registry & Model Deployment

- **imaging-ml:** Trains imaging classifiers, registers in ML Registry for SQL inference
- **survival-analysis:** Models registered for reproducible time-to-event analysis

7.3 Applications & Visualization

Streamlit in Snowflake

- Imaging dashboard, pharmacovigilance signals, claims analytics, clinical data explorer

Snowpark Container Services (SPCS)

- DICOM pixel viewer (OHIF/Cornerstone.js), deep learning inference on GPU, bioinformatics pipelines

7.4 Security & Governance

- **Sensitive Data Classification:** SYSTEM$CLASSIFY on all clinical tables for PHI auto-detection
- **Data Masking:** IS_ROLE_IN_SESSION() based masking for patient names, IDs, dates
- **Row-Access Policies:** Institutional data segregation by role
- **Audit Trails:** ACCESS_HISTORY monitoring for all PHI-containing tables

8. Cross-Domain Solution Patterns

The most powerful capability of the orchestrator is composing multiple skills for solutions that span healthcare business domains. Below are 9 pre-defined composition patterns.

Pattern 1: Imaging + Clinical Integration

- Step 1: $healthcare-imaging (dicom-parser) - Build imaging metadata tables
- Step 2: $fhir-data-transformation - Ingest FHIR DiagnosticReport/ImagingStudy
- Step 3: $clinical-nlp - Extract findings from radiology reports
- Step 4: PubMed CKE - Enrich with radiology research context
- Step 5: Platform: developing-with-streamlit OR build-react-app

Pattern 2: Clinical Data Warehouse (OMOP)

- Step 1: $fhir-data-transformation - Ingest FHIR bundles
- Step 2: $omop-cdm-modeling - Transform to OMOP CDM with vocabulary mapping
- Step 3: Platform: sensitive-data-classification, data-policy - HIPAA governance
- Step 4: Platform: semantic-view-optimization - Semantic views for analytics
- Step 5: Platform: developing-with-streamlit - Clinical dashboards

Pattern 3: Drug Safety Signal Detection

- Step 1: $pharmacovigilance - Load and analyze FAERS data
- Step 2: PubMed CKE - Search literature for drug-event associations
- Step 3: $clinical-nlp - Extract adverse events from narrative text
- Step 4: $claims-data-analysis - Correlate with claims-based utilization
- Step 5: Platform: developing-with-streamlit - Safety signal dashboard

Pattern 4: Genomics + Clinical Outcomes

- Step 1: $nextflow-development - Run nf-core pipeline on sequencing data
- Step 2: $variant-annotation - Annotate variants with ClinVar/gnomAD
- Step 3: $survival-analysis - Correlate variants with patient outcomes
- Step 4: Platform: machine-learning - Train predictive models

Pattern 5: Single-Cell Analysis Pipeline

- Step 1: $single-cell-rna-qc - QC and filter scRNA-seq data
- Step 2: $scvi-tools - Deep learning integration and batch correction
- Step 3: Platform: machine-learning - Register models in Snowflake ML Registry

Pattern 6: Real-World Evidence Study

- Step 1: $claims-data-analysis - Build cohorts from claims data
- Step 2: Clinical Trials CKE - Cross-reference with registered trials
- Step 3: $omop-cdm-modeling - Standardize to OMOP CDM
- Step 4: $survival-analysis - Time-to-event outcomes analysis
- Step 5: $clinical-nlp - Enrich with unstructured clinical data
- Step 6: PubMed CKE - Validate findings against published literature
- Step 7: Platform: developing-with-streamlit - Study results dashboard

Pattern 7: Clinical Trial Design

- Step 1: $scientific-problem-selection - Validate research problem
- Step 2: Clinical Trials CKE - Search for similar/competing trials
- Step 3: PubMed CKE - Review biomedical literature for evidence
- Step 4: $clinical-trial-protocol-skill - Generate protocol document
- Step 5: $survival-analysis - Power analysis and endpoint design
- Step 6: $claims-data-analysis - Feasibility analysis from claims data

Pattern 8: Lab Data Modernization

- Step 1: $instrument-data-to-allotrope - Standardize instrument outputs
- Step 2: Platform: dynamic-tables - Incremental pipeline for lab data
- Step 3: Platform: developing-with-streamlit - Lab analytics dashboard

Pattern 9: Clinical Data Application (React)

- Step 1: Domain skills - Prepare backend data (FHIR, OMOP, imaging, claims)
- Step 2: Platform: build-react-app - Build React/Next.js app with Snowflake data
- Step 3: Platform: deploy-to-spcs - Deploy containerized app to SPCS
- Step 4: Platform: data-policy - Enforce PHI masking at the API layer

9. Walkthrough: DICOM Pipeline (End-to-End)

This walkthrough shows how the cross-cutting prerequisite pattern works when a user asks: 'Build the entire DICOM ingestion to analytics pipeline.' The router auto-queries the data model knowledge repository before each sub-skill.

Step 0: Auto Pre-Step (Router)

The healthcare-imaging router detects INGEST + ANALYTICS intents. Before dispatching, it queries DICOM_MODEL_SEARCH_SVC:

```
SELECT SNOWFLAKE.CORTEX.SEARCH_PREVIEW(
    'UNSTRUCTURED_HEALTHDATA.DATA_MODEL_KNOWLEDGE.DICOM_MODEL_SEARCH_SVC',
    '{"query": "study series instance patient columns for ingestion",
      "columns": ["table_name","column_name","data_type","constraints",
                  "dicom_tag","relationships"]}'
);
```

The search results (table names, column definitions, DICOM tags, FK relationships) become the grounding context for subsequent steps.

Step 1: Create Schema ($healthcare-imaging -> dicom-parser)

The dicom-parser sub-skill receives the search results and uses them to generate DDL. Instead of hardcoded CREATE TABLE statements, it calls Cortex AI COMPLETE with the model reference:

```
SELECT SNOWFLAKE.CORTEX.COMPLETE('llama3.1-70b',
    'Generate Snowflake CREATE TABLE DDL from this reference: '
    || <search_results>::STRING
) AS generated_ddl;
```

Step 2: Build Ingestion Pipeline ($healthcare-imaging -> dicom-ingestion)

The dicom-ingestion sub-skill uses the same grounding context to build accurate COPY INTO column mappings and Dynamic Table SELECT lists with correct column names, types, and DICOM tag paths.

```
CREATE OR REPLACE DYNAMIC TABLE dicom_studies
  TARGET_LAG = '10 minutes'  WAREHOUSE = imaging_wh
AS
SELECT
  -- Column list grounded by DICOM_MODEL_SEARCH_SVC results
  metadata:StudyInstanceUID::STRING AS study_instance_uid,
  metadata:PatientID::STRING AS patient_id,
  ...
FROM dicom_raw;
```

Step 3: Build Analytics ($healthcare-imaging -> dicom-analytics)

Analytical Dynamic Tables reference the model-accurate column names from Step 0. PubMed CKE is available for research context enrichment.

Step 4: Apply Governance ($healthcare-imaging -> imaging-governance)

The governance sub-skill receives PHI column flags (contains_phi = 'Y') from the search results and auto-generates masking policies for every identified PHI column across all 18 tables.

```
-- PHI columns identified by Step 0 search results:
-- dicom_patient.patient_name, dicom_patient.patient_id,
-- dicom_patient.patient_birth_date, dicom_study.referring_physician,
-- dicom_equipment.device_serial_number, ...

ALTER TABLE dicom_patient MODIFY COLUMN patient_name
  SET MASKING POLICY phi_string_mask;
ALTER TABLE dicom_patient MODIFY COLUMN patient_id
  SET MASKING POLICY phi_string_mask;
-- ... (auto-generated for all PHI columns found)
```

10. Governance & Guardrails

All healthcare skills enforce these governance rules as cross-cutting concerns. The orchestrator profile embeds these as mandatory guardrails.

Guardrail	Implementation
HIPAA PHI Protection	All patient data tables require masking policies before analytics or dashboard exposure
Masking Policy Pattern	Always use IS_ROLE_IN_SESSION() (not CURRENT_ROLE()) for role-based masking
Audit Trails	ACCESS_HISTORY queries monitor all PHI access; templates in imaging-governance
De-identification	Prefer de-identified datasets for ML/analytics; use SHA2 hashing for UIDs
Data Classification	Run SYSTEM$CLASSIFY on all clinical tables to auto-detect PHI before pipelines
Data Model Grounding	Query DICOM_MODEL_SEARCH_SVC to identify PHI columns from reference model
Row-Access Policies	Restrict data by institution/department using role mapping tables
Data Quality	Always validate FHIR/HL7/OMOP data quality before building downstream tables
Genomic Data	Ensure proper consent tracking and data use agreements
FAERS Limitations	Always note limitations of spontaneous reporting data

11. Getting Started

Installation

1. Clone the skills repository:

```
git clone <repo-url> coco-healthcare-skills
```

2. Register skills in ~/.snowflake/cortex/skills.json. A ready-to-use template is included in the repo at skills.json.template -- just replace the path placeholder:

```
{
  "local": [{
    "path": "<ABSOLUTE_PATH_TO_REPO>/skills",
    "skills": [
      {"name": "healthcare-imaging",
       "relative_path": "medical-imaging/healthcare-imaging"},
      {"name": "claims-data-analysis",
       "relative_path": "claims-rwe/claims-data-analysis"},
      {"name": "clinical-nlp",
       "relative_path": "clinical-data-ehr/clinical-nlp"},
      {"name": "clinical-trial-protocol-skill",
       "relative_path": "drug-safety/clinical-trial-protocol-skill"},
      {"name": "fhir-data-transformation",
       "relative_path": "clinical-data-ehr/fhir-data-transformation"},
      {"name": "instrument-data-to-allotrope",
       "relative_path": "lab-instrument-data/instrument-data-to-allotrope"},
      {"name": "nextflow-development",
       "relative_path": "genomics-bioinformatics/nextflow-development"},
      {"name": "omop-cdm-modeling",
       "relative_path": "clinical-data-ehr/omop-cdm-modeling"},
      {"name": "pharmacovigilance",
       "relative_path": "drug-safety/pharmacovigilance"},
      {"name": "scientific-problem-selection",
       "relative_path": "research-strategy/scientific-problem-selection"},
      {"name": "scvi-tools",
       "relative_path": "genomics-bioinformatics/scvi-tools"},
      {"name": "single-cell-rna-qc",
       "relative_path": "genomics-bioinformatics/single-cell-rna-qc"},
      {"name": "survival-analysis",
       "relative_path": "genomics-bioinformatics/survival-analysis"},
      {"name": "variant-annotation",
       "relative_path": "genomics-bioinformatics/variant-annotation"}
    ]
  }]
}
```

3. Copy the agent profile:

```
cp coco-healthcare-skills/agents/healthcare-solutions.md \
    ~/.snowflake/cortex/agents/
```

4. Set up CKEs (optional): Install PubMed CKE and/or Clinical Trials CKE from Snowflake Marketplace.

5. Set up Data Model Knowledge Repository: Run scripts/setup_dicom_model_knowledge_repo.sql to create the Cortex Search Service.

6. Update skills (git pull): Since all skills are registered from the repo path, running git pull updates every skill automatically.

7. Verify in Cortex Code:

```
/agents   # Should show healthcare-solutions
/skill    # Should show all healthcare skills
```

Usage Examples

```
# Invoke the orchestrator for any healthcare task:
$healthcare-imaging parse these DICOM files from S3
$fhir-data-transformation load Patient and Observation bundles
$pharmacovigilance analyze FAERS data for aspirin adverse events
$claims-data-analysis build a T2D cohort from claims
$variant-annotation annotate this VCF with ClinVar
$survival-analysis run KM analysis on treatment outcomes
```

File Locations

Component	Path
Agent Profile	~/.snowflake/cortex/agents/healthcare-solutions.md
Skills Config	~/.snowflake/cortex/skills.json
Skills Config Template	coco-healthcare-skills/skills.json.template
Skills (repo root)	coco-healthcare-skills/skills/
medical-imaging/	healthcare-imaging (router + 7 sub-skills)
clinical-data-ehr/	fhir-data-transformation, clinical-nlp, omop-cdm-modeling
drug-safety/	pharmacovigilance, clinical-trial-protocol-skill
claims-rwe/	claims-data-analysis
genomics-bioinformatics/	nextflow-development, variant-annotation, single-cell-rna-qc, scvi-tools, survival-analysis
lab-instrument-data/	instrument-data-to-allotrope
research-strategy/	scientific-problem-selection
Reference Model	references/dicom_data_model_reference.xlsx
Setup SQL	scripts/setup_dicom_model_knowledge_repo.sql

Snowflake Objects

Object	Fully Qualified Name
Data Model Table	UNSTRUCTURED_HEALTHDATA.DATA_MODEL_KNOWLEDGE.DICOM_MODEL_REFERENCE
Cortex Search Service	UNSTRUCTURED_HEALTHDATA.DATA_MODEL_KNOWLEDGE.DICOM_MODEL_SEARCH_SVC
Stage	UNSTRUCTURED_HEALTHDATA.DATA_MODEL_KNOWLEDGE.dicom_model_stage